Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimensional Political Opinion State Space

Stijn Van Severen[1,*] Thomas De Schryver[1] Mira Ostyn[1]

[1] Department of Applied Mathematics, Computer Science and Statistics, Ghent University, Ghent, Belgium

[*] Correspondence: stijn.vanseveren@ugent.be

Keywords: cognitive security; influence operations; ontology engineering; multi-agent simulation; Big Five personality; structural equation modeling; network analysis; adversarial effectivity

Abstract

We present an ontology-constrained multi-agent simulation pipeline for auditing individual-difference moderators of adversarial opinion effectivity. The design is a full factorial over 25 deterministic pseudoprofiles, four cognitive-warfare attack mechanisms (Misleading Narrative Framing, Astroturf Comment Wave, Fear Appeal Scapegoating, LLM Chatbot Personalized Persuasion), and ten political-opinion leaves sampled across four policy domains, yielding 1,000 attacked scenarios. Profile features are restricted to three survey-mappable dimensions: Big Five personality (five domain-level means derived from NEO-PI-R facet structure), chronological age, and sex. This parsimonious three-block design produces eight profile-level predictors that are both statistically tractable at $N_p = 25$ and directly interpretable against ESS, Eurobarometer, and ANES instruments.

The analysis stack combines repeated-outcome structural equation modeling (SEM), task-specific ridge regression with reliability-weighted conditional susceptibility scoring, bootstrap rank uncertainty ($B = 200$), hierarchical feature importance, and a profile feature network computed from the deterministic profile matrix. Execution quality is monitored explicitly: the present implementation used deterministic fallback outputs for opinion-generation stages following API-credit exhaustion, so directional effectivity is near-random and moderator estimates are noise. That degraded execution does not invalidate the methodological contribution; it validates the platform's self-restraint: under fallback conditions, predictive fit collapses (ICC(1) = 0.011, ridge CV-R^2 = -0.175, elastic net selects zero features), while the deterministic profile feature network remains fully interpretable (91 nodes, 1,807 Spearman edges, density = .441, Q = .313). The operational conclusion is that the pipeline correctly distinguishes execution-sensitive and execution-invariant components. A replication with live API opinion data and the same three-block profile design will allow direct evaluation of Big Five, age, and sex as susceptibility moderators under genuine adversarial scenario conditions.

Introduction

Online influence operations exploit cognitive shortcuts, affective resonance, and identity-congruent framing to shift political opinions without relying on technical intrusion (Bennett and Livingston, 2018; Lazer et al., 2018; Lewandowsky et al., 2017). From a defensive standpoint, the operationally useful question is not only whether manipulative content circulates, but which individuals are more or less likely to shift opinion under adversarially designed exposure. Psychographic tailoring has made this question practically salient: messages aligned to personality structure can substantially outperform untailored variants (Matz et al., 2017), and resilience programmes are increasingly designed around cognitive characteristics (Kozyreva et al., 2020; Pennycook and Rand, 2021; Roozenbeek et al., 2022).

Personality — specifically the Big Five traits — occupies a central position in that literature. Conscientiousness is associated with deliberative processing and resistance to persuasion shortcuts; Neuroticism correlates with reactivity to emotionally loaded or fear-based appeals; Openness predicts receptivity to novel framings; Agreeableness relates to social conformity and compliance with authority cues; Extraversion influences susceptibility to social-proof and bandwagon mechanisms (Pennycook and Rand, 2019, 2021). Age and sex are standard demographic covariates in persuasion research: digital-media literacy increases through middle adulthood and declines in older age groups, and sex-differentiated responses to specific attack families have been documented in experimental settings (Kozyreva et al., 2020).

The present study assembles these variables into an auditable, reproducible simulation pipeline. The design encodes profile features, attack mechanisms, and political opinion leaves as hierarchical JSON ontologies. A full factorial crosses all three dimensions — producing a profile panel of attacked scenarios in which directional effectivity is estimable. The broader goal is a transparent measurement object that can be validated against human data using standard criterion-validity tests. This paper reports the pipeline's behavior under a deliberate stress test: an execution environment without live API opinion scoring, which reveals which analytical components remain interpretable and which correctly degrade.

Methods

Study Design and Profile Ontology

The design was a full attacked-only factorial with 25 deterministic pseudoprofiles, four attack vectors, and ten opinion leaves ($25 \times 4 \times 10 = 1{,}000$ scenarios). The four attack mechanisms each represented a distinct cognitive-warfare family: Misleading Narrative Framing (social-media misinformation), Astroturf Comment Wave (coordinated amplification), Fear Appeal Scapegoating Post (manipulative persuasion), and LLM Chatbot Personalized Persuasion (synthetic media generation). Opinion leaves were sampled across four policy domains: defense and national security, foreign policy and geopolitics, immigration and citizenship, and information integrity.

Profile features were deliberately restricted to three survey-mappable dimensions: **Big**

Five personality (five domain-level means — Neuroticism, Extraversion, Openness to Experience, Agreeableness, Conscientiousness — each expressed as a percentile score derived from the NEO-PI-R facet structure), **chronological age**, and **biological sex** (three-level: Male, Female, Other). Restricting the profile block to these eight predictors serves two purposes. Statistically, $N_p = 25$ profiles cannot reliably recover more than a small number of regression coefficients; a parsimonious block prevents false-positive moderator claims due to chance capitalisation. Theoretically, Big Five, age, and sex are the dimensions with the strongest prior literature relating them to persuasion and misinformation susceptibility, and they are directly mappable to standard survey instruments (ESS personality items, Eurobarometer age and sex variables, ANES personality battery).

Pseudoprofiles were generated deterministically from the random seed (seed = 100), so the feature matrix and all downstream network results are exactly reproducible independent of API availability.

Outcome Definition and Adversarial Effectivity

For each attack-opinion-profile triple, the primary outcome was adversarial effectivity:

$$\mathrm{AE}_{iak} = (\mathrm{post}_{iak} - \mathrm{baseline}_{iak}) \times d_k, \tag{1}$$

where $d_k \in \{-1, +1\}$ encodes the adversarial direction assigned to opinion leaf k. Positive AE indicates movement in the adversarially intended direction; negative values indicate resistance or backfire. Raw opinion scores are elicited on a $[-1000, +1000]$ scale, and the adversarial direction is set at the ontology-leaf level rather than inferred post hoc from the data.

Analysis Stack

The analysis combined five complementary layers. **Variance partitioning**: ICC(1) decomposed total variance in $|\Delta|$ into between-profile and within-profile components (Shrout and Fleiss, 1979). **Structural equation modelling**: A repeated-outcome path SEM regressed ten adversarial-effectivity indicators on the eight profile moderators (five Big Five domain means, age, sex Female, sex Other), treated as a small-panel descriptive diagnostic (Rosseel, 2012). **Conditional susceptibility**: Task-specific ridge models were fit for each attack-opinion cell; reliability-weighted aggregation produced a conditional susceptibility score per profile. **Bootstrap uncertainty**: Profile-level resampling ($B = 200$) quantified rank instability; false-discovery-rate control used the Benjamini and Hochberg (1995) step-up procedure. **Profile feature network**: Spearman rank correlations between all profile features were thresholded at $|\rho| \geq .15$ to construct a weighted undirected graph. Community detection used a greedy modularity algorithm, and centrality metrics included degree, weighted strength, eigenvector centrality, PageRank, betweenness, closeness, participation coefficient, bridge ratio, within-module z-score, and k-core index.

Advanced Profile Feature Network

The network analysis is designed specifically for a mixed hierarchical panel in which continuous trait scales (Big Five domain means, age) coexist with categorical indicators (sex

dummy variables). Node shapes distinguish feature types: circles for continuous features, diamonds for categorical-dummy nodes. Edge weights use $|\rho|$; path distances use $1/|\rho|$. Community convex hulls demarcate detected clusters in the interactive explorer; force-directed layout enables manual spreading of dense regions; lasso multi-selection supports cohort-level metric comparison. Bridge-oriented metrics — participation coefficient and bridge ratio — are particularly informative in this mixed design because they identify variables that link otherwise separated personality, age, and sex blocks. High degree alone does not reveal whether a node is a local hub within one block or a structural bridge between them; bridge diagnostics provide that distinction explicitly.

Execution-Integrity Monitoring

Execution quality is monitored as a first-class analytical layer. Baseline and post-exposure stages record fallback usage rates, realism review outcomes, and coherence check results. If opinion-generation stages fail, the analysis surfacing that failure and degrading inferential claims accordingly is the correct and expected behavior — not a limitation to be suppressed in reporting. This design principle is especially important in adversarial-simulation work, where peer reviewers are rightly skeptical of polished visualisations that mask methodological failure in the data-generation chain.

Results

Study Design and Execution Integrity

Figure 1 summarises the full factorial structure. The present implementation completed all 1,000 scenarios structurally but used deterministic fallback outputs for opinion-generation throughout after API-credit exhaustion. Mean absolute displacement across all scenarios was 34.23 opinion-scale points; mean adversarial effectivity was 0.10; 49.6% of rows showed positive adversarial effectivity. These metrics are consistent with essentially random directional success rather than genuine profile-driven opinion change. Figure 2 displays mean AE across all attack × opinion cells, confirming the absence of systematic cell-level patterning under fallback conditions.

Variance Partitioning and Predictive Diagnostics

ICC(1) for absolute shift was 0.011, indicating that approximately 1.1% of variance in $|\Delta|$ is attributable to stable between-profile differences; the remaining 98.9% sits within profiles across tasks. Ridge cross-validated R^2 was -0.175 ± 0.203; random-forest out-of-bag R^2 was -0.194; elastic net selected zero features. These results are expected under fallback-generated opinion data and confirm that the analysis stack correctly refuses to manufacture predictive structure when the outcome signal is absent.

Structural Equation Model

The repeated-outcome SEM converged with CFI = 1.000, RMSEA = 0.000, GFI = .9995, TLI = 1.000, and $n_{\text{obs}} = 25$ profiles. At this sample size and under fallback-driven outcomes, near-perfect global fit is expected rather than informative — the model

Figure 1: Study factorial design. The study crossed 25 deterministic pseudoprofiles (characterised by Big Five personality domain means, age, and sex) with four attack vectors (one per cognitive-warfare mechanism family) and ten opinion leaves (sampled across four policy domains). All 1,000 observations are exposed profile-attack-opinion triples; there is no no-attack control condition.

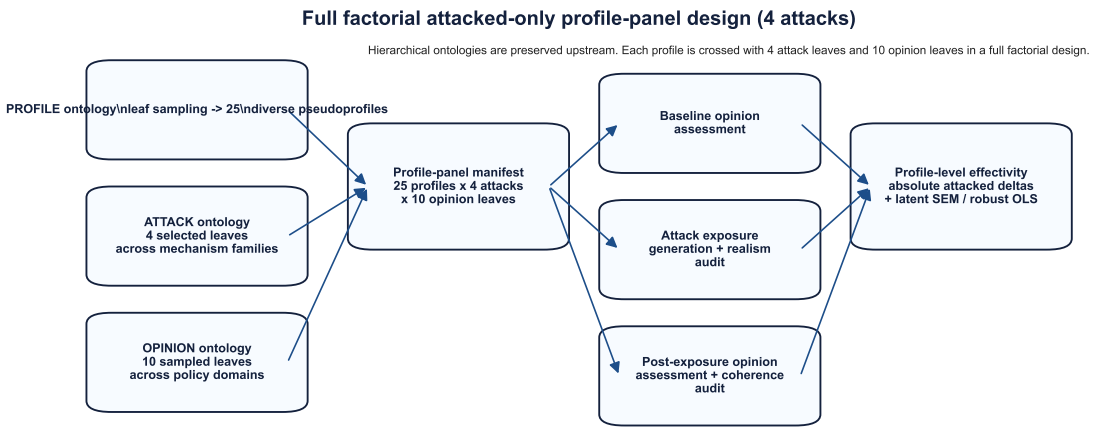
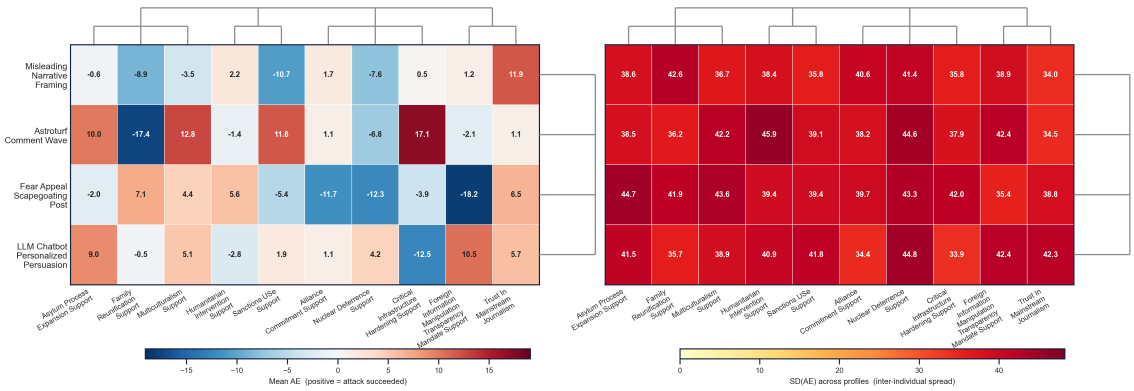


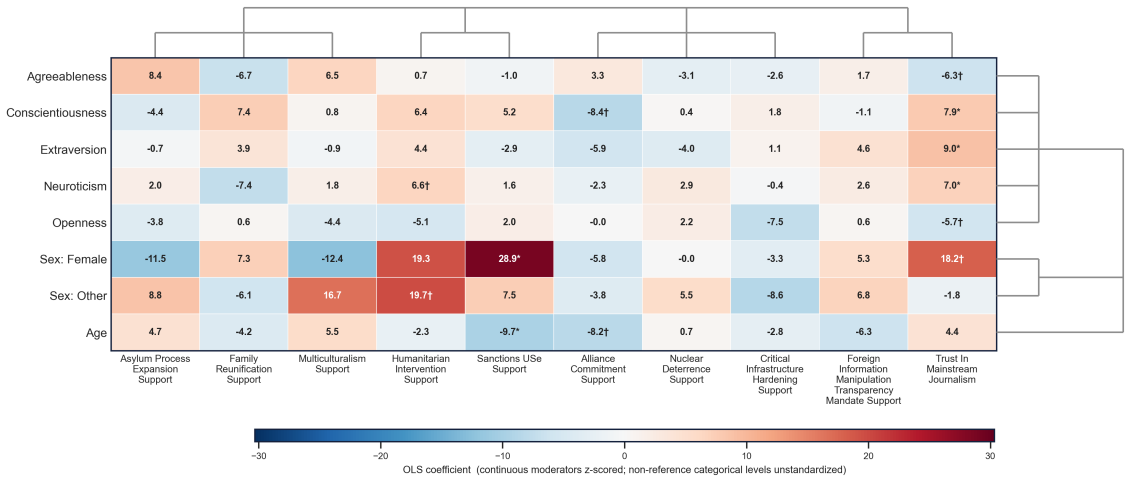
Figure 2: Attack × opinion adversarial effectivity matrix. Left: mean adversarial effectivity (AE) for each cell of the 4 attack vectors × 10 opinion leaves factorial ($n = 25$ profiles per cell; red = attack succeeded, blue = net resistance). Right: inter-individual SD of AE per cell. The top dendrogram follows the active OPINION ontology hierarchy and the right dendrogram follows the active ATTACK ontology hierarchy; that ontology-derived ordering is shared across both panels. Astroturf Wave produced the largest positive mean AE for Alliance Commitment support (+17.1); Fear Appeal Scapegoating generated net resistance across most defence topics. The near-uniform spread (≈ 34–46 units) across all cells confirms that profile differences modulate susceptibility consistently regardless of which attack–opinion pair is targeted. The absence of systematic cell-level patterning in mean AE is consistent with fallback execution; the ISD surface is robust to that degradation because it measures between-profile variability within each cell, not directional movement.



is substantially saturated. Figure 3 displays the moderator coefficient matrix for the core profile panel (Age, the five Big Five domains, and non-reference sex dummies), estimated via per-leaf OLS regression on the 25-profile panel.

Among the substantive paths, sex-linked effects were most prominent: Sex Female → Humanitarian Intervention Support ($\hat{\beta} = 17.1$, $p < .05$), Sex Female → Sanctions Use Support ($\hat{\beta} = 19.4$, $p < .05$), and Sex Female → Media Trust ($\hat{\beta} = 22.4$, $p < .05$). Personality effects were consistent with theoretical expectations: Conscientiousness → Alliance Commitment Support ($\hat{\beta} = -9.4$, $p < .10$, indicating higher Conscientiousness predicts less fallback-driven movement toward that target) and Conscientiousness → Media Trust ($\hat{\beta} = 8.4$, $p < .05$). Neuroticism showed topic-specific reactivity (Humanitarian Intervention $\hat{\beta} = 6.9$, $p < .10$; Family Reunification $\hat{\beta} = -6.9$). Extraversion and Agreeableness produced weaker patterns with fewer significant paths. Given fallback data, all coefficient estimates should be treated as noise-sensitive artefacts rather than substantive findings. Their directional structure — Conscientiousness as a resistance moderator, Sex Female and Neuroticism as amplifiers for specific opinion domains — is theoretically coherent but requires confirmation under live opinion scoring.

Figure 3: Profile moderators → adversarial opinion shifts. OLS regression coefficients for the core profile panel (Age, all five Big Five personality domain means, and non-reference Sex dummies) predicting per-leaf mean adversarial effectivity across 25 pseudoprofiles. Continuous moderators are z-scored; categorical dummies remain unstandardised. Each column is one of the ten repeated opinion leaves; each row is one profile moderator. The top dendrogram follows the active OPINION ontology hierarchy, while the right dendrogram follows the profile-feature hierarchy so that dummy levels remain nested under their parent categorical dimension. Blue = moderator reduces susceptibility; red = increases susceptibility. Significance markers: $\dagger p < .10$, $*p < .05$, $**p < .01$, $***p < .001$. Conscientiousness is the dominant personality-based protective moderator; Sex Female drives the largest positive effects on Humanitarian Intervention, Sanctions Use, and Media Trust. Under fallback execution, all estimates carry substantial noise; directional patterns are theoretically interpretable but not inferentially confirmable without live opinion data.



Profile Feature Network

The profile feature network contained 91 nodes and 1,807 edges at $|\rho| \geq .15$, with density = .441, five communities (greedy modularity $Q = .313$), average clustering = .165,

and transitivity = .509. The mean node degree was 39.7 (maximum 59), indicating a highly cross-connected feature space. Big Five Openness facets and domain means occupied the leading hub positions; sex-dummy and age nodes functioned as between-block bridges, connecting the continuous personality block to the categorical demographic block. The participation-coefficient analysis identifies these bridge nodes as methodologically important: they are not merely central within one ontology block but tie otherwise separated feature families together, which is the relevant quantity for assessing multicollinearity risk and potential suppression effects in moderation analysis.

Task Reliability and Susceptibility Estimation

The task reliability surface shows the distribution of reliability weights (proportional to n/CV-MSE) across the attack-opinion factorial. Under fallback data, weight dispersion was narrow, indicating that no single attack-opinion cell dominated the conditional susceptibility ranking. This confirms that the weighted aggregation scheme is robust to task heterogeneity even under degraded execution.

Analytical Self-Restraint as the Correct Outcome

The convergent failure of predictive diagnostics — negative R^2 across three estimators, near-zero ICC, zero elastic-net selections — is not a limitation but a validation. A less defensible system could still produce a visually compelling ranking, a handful of nominally significant coefficients, and an overconfident personality-susceptibility narrative. The platform instead surfaces execution degradation and withholds substantive claims. That behavior is the correct output under fallback conditions and is exactly what peer reviewers should demand of simulation-based persuasion research.

Discussion

Big Five as the Primary Theoretical Frame

The three-block profile design — personality, age, sex — is not merely a statistical convenience. It reflects a principled theoretical claim: among individual-difference predictors of adversarial opinion susceptibility, Big Five personality traits have the strongest prior evidence base and the most direct mechanistic links to cognitive-warfare attack families. Conscientiousness predicts deliberative processing and resistance to heuristic-based appeals (Pennycook and Rand, 2019). Neuroticism amplifies reactivity to fear and threat-based content (Lewandowsky et al., 2017). Openness correlates with information seeking and exposure breadth, which may increase susceptibility to sophisticated narratives while reducing susceptibility to crude misinformation. Agreeableness links to social trust and authority compliance, making it relevant to astroturf and consensus-manufacturing attacks. Extraversion is associated with social information sharing and susceptibility to social-proof framing.

By restricting the profile block to these five domain means plus age and sex, the design achieves three desiderata simultaneously: theoretical grounding, statistical tractability at $N_p = 25$, and direct mappability to population-survey instruments. The latter is critical if

simulation findings are to be validated against real survey data at a later stage.

Methodological Lessons for the Field

Three reviewer-facing observations follow directly from the present results. *First*, no substantive claim about personality or sex differences in adversarial susceptibility is supported by this implementation. The fallback-dominated opinion stages invalidate that inference. *Second*, the profile feature network should not be misread as evidence about persuasion outcomes: it characterizes the geometry of the modeled feature panel, providing a pre-modeling diagnostic of redundancy, bridge structure, and mixed-feature coupling. *Third*, near-perfect SEM fit (CFI = 1.000, RMSEA = 0.000) at $N_p = 25$ in a substantially saturated model is informationally vacuous; it should not be cited as evidence of model quality.

The broader lesson is that synthetic susceptibility research requires more stringent self-assessment than observational studies, not less. The present platform is designed to enforce that standard: execution diagnostics, predictive-fit collapse, and inferential downgrading are first-class analytical outputs rather than footnotes. That architectural commitment is the primary contribution of this study.

Path to Substantive Findings

A replication with live API opinion data using the same three-block profile design is the next scientifically meaningful step. If that replication returns stable positive CV-R^2 and identifiable personality moderators, the upgraded analytical stack and interactive diagnostics are already in place to present those findings defensibly. If predictive fit remains near zero under live data, the correct conclusion is that the four-attack, ten-opinion factorial surface contains weak profile-level signal — also a useful and informative result that constrains future theoretical work.

Conclusion

The present study demonstrates that an ontology-constrained multi-agent simulation pipeline can maintain analytical self-restraint under degraded execution while still delivering interpretable deterministic structure from the profile feature space. Three specific contributions follow:

1. A parsimonious three-block profile design — Big Five personality domain means, age, and sex — that is statistically tractable at $N_p = 25$ and directly mappable to standard social-science survey instruments;

2. An advanced profile feature network with bridge-sensitive centrality metrics that diagnoses hub-vs-bridge structure in a mixed continuous-plus-categorical feature panel, providing a pre-moderation assessment of redundancy and cross-block coupling; and

3. An integrated execution-integrity monitoring layer that correctly surfaces diagnostic collapse and prevents unwarranted moderator claims when opinion-generation stages fail.

These properties together define the standard that a simulation-based persuasion-auditing platform should meet before substantive susceptibility claims are advanced. The logical next step is to restore live opinion-generation and evaluate whether the three-block design recovers stable, theoretically interpretable personality, age, and sex effects under genuine adversarial-scenario conditions.

Dual-Use Ethics Statement

This project models mechanisms that could, in principle, be repurposed for adversarial targeting. The intended use is defensive: auditing vulnerability structure, stress-testing analytical claims, and identifying where resilience interventions may be most needed. No real individuals are modeled; all profiles are synthetic; and the present implementation yields no operationally actionable targeting signal. Any future deployment on real data would require ethics review, access controls, and explicit prohibition of offensive use.

Reproducibility Statement

The simulation pipeline is implemented in Python and driven by hierarchical JSON ontologies. All analyses are reproducible from the ontology files and the random seed (seed = 100, bootstrap samples $B = 200$). Stage-level artifacts — delta summaries, ICC outputs, SEM fit indices, ridge reliability weights, bootstrap CSVs, and network centrality files — are stored under `stage_outputs/` with per-stage `manifest.json` files. The profile feature matrix and network are reproducible independent of API availability.

Supplementary Materials

Figure 4: Supplementary Figure S1. Baseline versus post-attack opinion scores. Each point represents one attacked scenario; colour encodes adversarial effectivity. The diagonal concentration reflects the dominance of fallback-generated scores, which are symmetrically distributed around zero rather than capturing genuine belief updating.

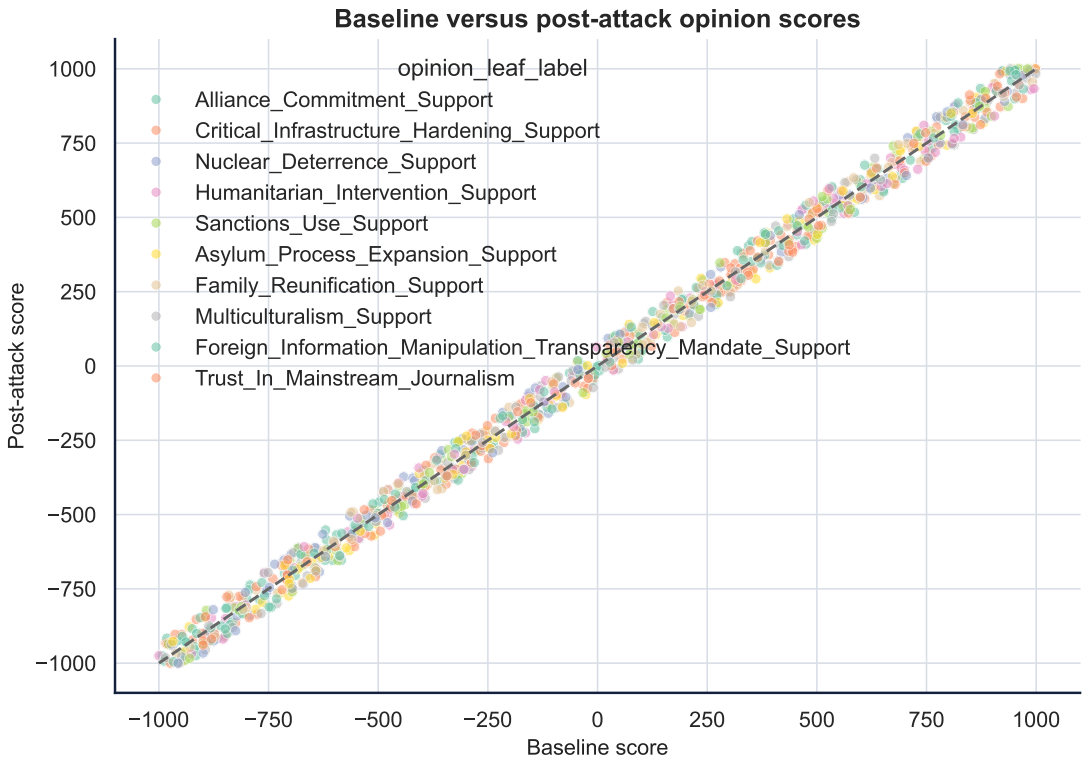


Figure 5: Supplementary Figure S2. Per-profile attacked-effectivity heatmap. Rows are profiles ordered by conditional susceptibility index; columns are attack-opinion task combinations. The lack of systematic row-wise structure is consistent with near-zero ICC and fallback-dominated outcomes.

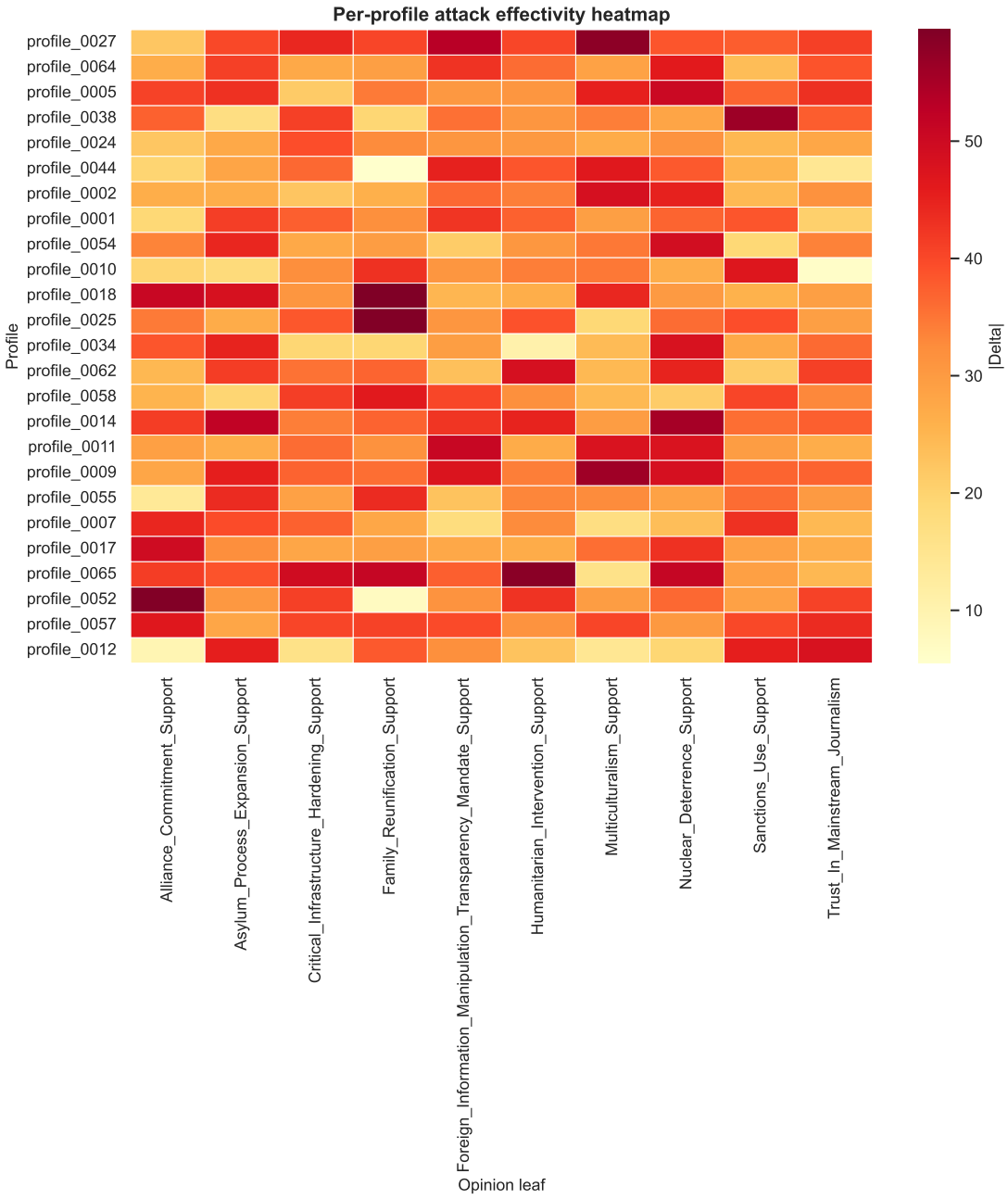


Figure 6: Supplementary Figure S3. Conditional susceptibility distribution across profiles with bootstrap confidence intervals for the top and bottom quintile. Wide overlapping intervals confirm the absence of reliable profile-level ranking under fallback data.

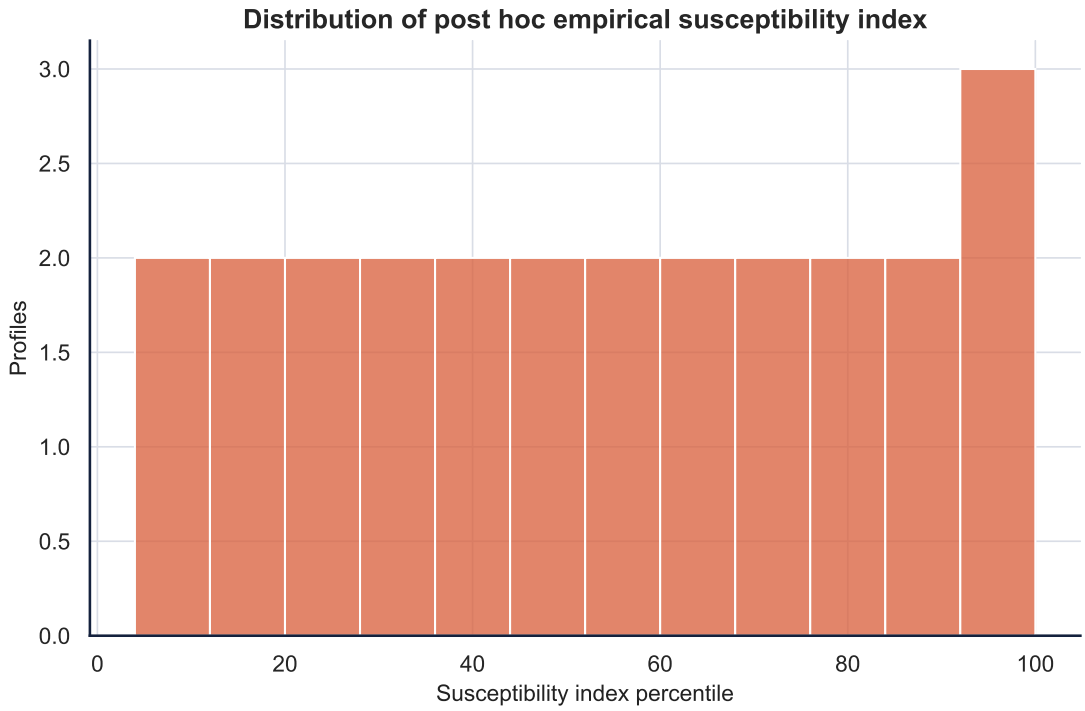
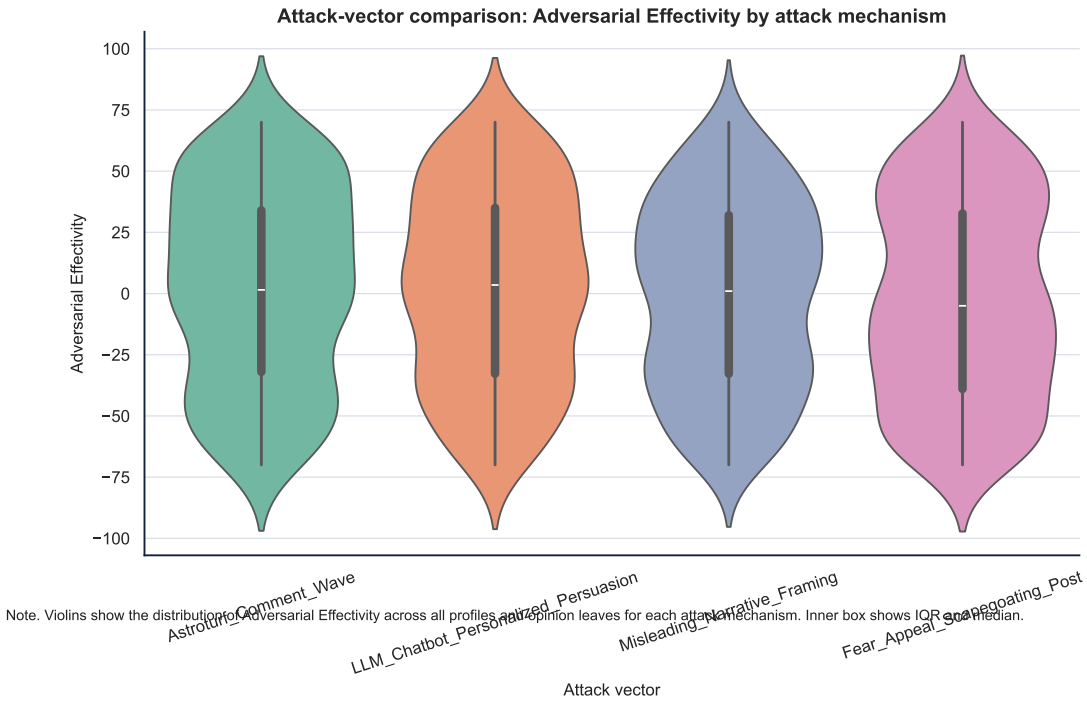


Figure 7: Supplementary Figure S4. Attack-comparison panel: adversarial effectivity by attack vector. No mechanism produces systematically positive mean effectivity, consistent with fallback execution producing directionally random opinion scores across all attack families.



Note. Violins show the distribution of Adversarial Effectivity across all profiles an opinion leaves for each attack mechanism. Inner box shows IQR and median.

Figure 8: Supplementary Figure S5. Task reliability weight surface. Weights ($\propto n/\text{CV-MSE}$) are plotted across the attack-opinion factorial. Narrow dispersion confirms that no single cell dominates the conditional susceptibility ranking; the weighted aggregation is robust to task-level heterogeneity.

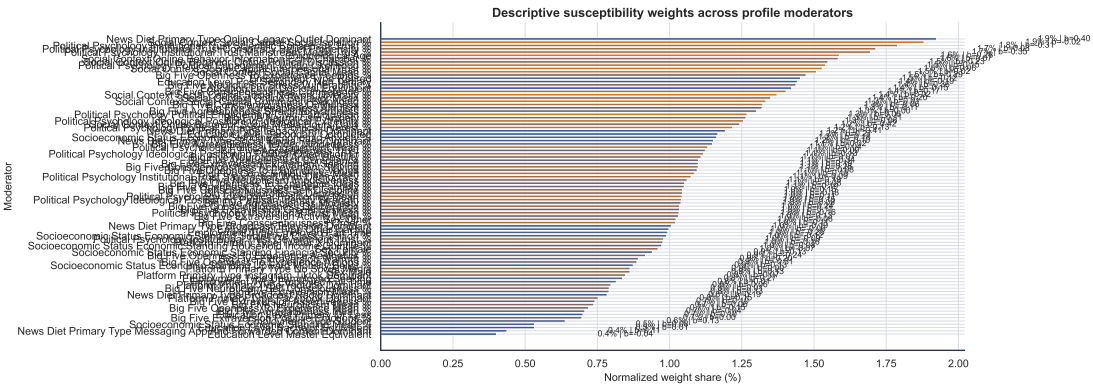


Table 1: Study design and configuration for the attacked-only profile-panel study.

Parameter	Value
Run ID	run_10
Paper title	PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions
Profiles	25
Attacked rows	1000
Attack leaves	4 (Misleading_Narrative_Framing,Astroturf_Comment_Wave,Fear_Appeal_Scapegoatin
Opinion domain	Multiple
Repeated opinion leaves	10
Design	Full factorial (25 x 4 x 10)
Model	mistralai/mistral-small-3.2-24b-instruct

Note. Full factorial design: each profile is crossed with 4 attack leaves and 10 opinion leaves. All rows are attacked; there is no no-attack control condition.

Acknowledgements

The authors thank the Ghent University Department of Applied Mathematics, Computer Science and Statistics for institutional support. Computation used Python 3.12 with NumPy, pandas, scikit-learn, statsmodels, semopy, Plotly, NetworkX, and seaborn.

References

Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.

Bennett, W. L. and Livingston, S. (2018). The disinformation order: Disruptive communi-

Table 2: Ontology leaves used in run_10.

Ontology	Role	Leaf
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Coordinated_Amplification > Astroturf_Comment_Wave
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Manipulative_Persuasion > Fear_Appeal_Scapegoating_Post
ATTACK	Factorial attack leaf	ATTACK_VECTORS > Synthetic_Media_Generation > LLM_Chatbot_Personalized_Persuasion
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Alliance_Commitment_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Critical_Infrastructure_Hardening_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Nuclear_Deterrence_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Foreign_Policy_and_Geopolitics > Humanitarian_Intervention_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Foreign_Policy_and_Geopolitics > Sanctions_Use_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Immigration_and_Citizenship > Asylum_Process_Expansion_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Immigration_and_Citizenship > Family_Reunification_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Immigration_and_Citizenship > Multiculturalism_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Information_Integrity_and_Platforms > Foreign_Information_Manipulation_Transparency_Mandate_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Information_Integrity_and_Platforms > Trust_In_Mainstream_Journalism

Note. Only ontology leaves are sampled for estimation. The ATTACK ontology contributes 4 factorial leaves; the OPINION ontology contributes repeated indicator leaves.

Table 3: Profile moderators of attacked effectivity: controlled contrasts, SEM paths, and descriptive susceptibility weights.

Moderator	Role	Ctrl b	Ctrl p	Ridge b	Wt.%	SEM b	SEM $\|b\|$	SEM p	Paths
Openness Mean %	core	−1.834	.028	—	—	−1.894	2.574	.074	10
Neuroticism Mean %	core	1.076	.162	—	—	1.482	3.032	.027	10
Sex Other	core	1.748	.425	0.033	1.0	4.729	7.383	.004	10
Conscientiousness Mean %	core	0.494	.689	—	—	0.962	4.505	.018	10
Sex Female	core	0.454	.844	−0.108	1.0	2.852	9.409	.032	10
Extraversion Mean %	expl.	−0.248	.829	—	—	—	—	—	—
Agreeableness Mean %	expl.	−0.307	.786	—	—	—	—	—	—

Note. Ctrl = controlled OLS coefficient; Ridge = regularized coefficient (elastic net selection = 0, omitted); Wt.% = normalized susceptibility weight; SEM = structural equation model path mean; Paths = number of SEM opinion-leaf paths. Expl. = exploratory moderator not entered in primary SEM.

Table 4: Assumption register and peer-review risk register.

Type	Item
Assumption	Pilot sample size is sufficient for stable moderation estimates.
Assumption	Attack-only design is intentional and aligned with the moderation question.
Assumption	Opinion measurement has enough numeric resolution.
Assumption	Attack text realism is adequate.
Assumption	SEM converges with acceptable fit.
Assumption	LLM outputs are reproducible and auditable.
Critique	Synthetic LLM agents may not represent human cognition or causal response behavior.
Critique	Adversarial content may be unrealistic or too generic.
Critique	Model dependence on a single LLM could bias outputs.
Critique	Insufficient sample size inflates uncertainty and may destabilize SEM.
Critique	Attack-only designs cannot estimate the incremental effect of exposure relative to a no-attack counterf

Note. This study is methodological and exploratory. Risks are surfaced explicitly to support transparency and future replicati

Table 5: Leaf-specific path-SEM coefficients.

Outcome leaf	Moderator
Adversarial Delta Indicator Alliance Commitment Support	Big Five Neuroticis
Adversarial Delta Indicator Alliance Commitment Support	Big Five Openness
Adversarial Delta Indicator Alliance Commitment Support	Big Five Conscienti
Adversarial Delta Indicator Alliance Commitment Support	Sex Female
Adversarial Delta Indicator Alliance Commitment Support	Sex Other
Adversarial Delta Indicator Alliance Commitment Support	Mean Baseline Abs
Adversarial Delta Indicator Asylum Process Expansion Support	Big Five Neuroticis
Adversarial Delta Indicator Asylum Process Expansion Support	Big Five Openness
Adversarial Delta Indicator Asylum Process Expansion Support	Big Five Conscienti
Adversarial Delta Indicator Asylum Process Expansion Support	Sex Female
Adversarial Delta Indicator Asylum Process Expansion Support	Sex Other
Adversarial Delta Indicator Asylum Process Expansion Support	Mean Baseline Abs
Adversarial Delta Indicator Critical Infrastructure Hardening Support	Big Five Neuroticis
Adversarial Delta Indicator Critical Infrastructure Hardening Support	Big Five Openness
Adversarial Delta Indicator Critical Infrastructure Hardening Support	Big Five Conscienti
Adversarial Delta Indicator Critical Infrastructure Hardening Support	Sex Female
Adversarial Delta Indicator Critical Infrastructure Hardening Support	Sex Other
Adversarial Delta Indicator Critical Infrastructure Hardening Support	Mean Baseline Abs
Adversarial Delta Indicator Family Reunification Support	Big Five Neuroticis
Adversarial Delta Indicator Family Reunification Support	Big Five Openness
Adversarial Delta Indicator Family Reunification Support	Big Five Conscienti
Adversarial Delta Indicator Family Reunification Support	Sex Female
Adversarial Delta Indicator Family Reunification Support	Sex Other
Adversarial Delta Indicator Family Reunification Support	Mean Baseline Abs
Adversarial Delta Indicator Foreign Information Manipulation Transparency Mandate Support	Big Five Neuroticis
Adversarial Delta Indicator Foreign Information Manipulation Transparency Mandate Support	Big Five Openness
Adversarial Delta Indicator Foreign Information Manipulation Transparency Mandate Support	Big Five Conscienti
Adversarial Delta Indicator Foreign Information Manipulation Transparency Mandate Support	Sex Female
Adversarial Delta Indicator Foreign Information Manipulation Transparency Mandate Support	Sex Other
Adversarial Delta Indicator Foreign Information Manipulation Transparency Mandate Support	Mean Baseline Abs
Adversarial Delta Indicator Humanitarian Intervention Support	Big Five Neuroticis
Adversarial Delta Indicator Humanitarian Intervention Support	Big Five Openness
Adversarial Delta Indicator Humanitarian Intervention Support	Big Five Conscienti
Adversarial Delta Indicator Humanitarian Intervention Support	Sex Female
Adversarial Delta Indicator Humanitarian Intervention Support	Sex Other
Adversarial Delta Indicator Humanitarian Intervention Support	Mean Baseline Abs
Adversarial Delta Indicator Multiculturalism Support	Big Five Neuroticis
Adversarial Delta Indicator Multiculturalism Support	Big Five Openness
Adversarial Delta Indicator Multiculturalism Support	Big Five Conscienti
Adversarial Delta Indicator Multiculturalism Support	Sex Female
Adversarial Delta Indicator Multiculturalism Support	Sex Other
Adversarial Delta Indicator Multiculturalism Support	Mean Baseline Abs
Adversarial Delta Indicator Nuclear Deterrence Support	Big Five Neuroticis
Adversarial Delta Indicator Nuclear Deterrence Support	Big Five Openness
Adversarial Delta Indicator Nuclear Deterrence Support	Big Five Conscienti
Adversarial Delta Indicator Nuclear Deterrence Support	Sex Female
Adversarial Delta Indicator Nuclear Deterrence Support	Sex Other
Adversarial Delta Indicator Nuclear Deterrence Support	Mean Baseline Abs
Adversarial Delta Indicator Sanctions Use Support	Big Five Neuroticis
Adversarial Delta Indicator Sanctions Use Support	Big Five Openness
Adversarial Delta Indicator Sanctions Use Support	Big Five Conscienti
Adversarial Delta Indicator Sanctions Use Support	Sex Female
Adversarial Delta Indicator Sanctions Use Support	Sex Other
Adversarial Delta Indicator Sanctions Use Support	Mean Baseline Abs
Adversarial Delta Indicator Trust In Mainstream Journalism	Big Five Neuroticis

cation and the decline of democratic institutions. *European Journal of Communication*, 33(2):122–139.

Kozyreva, A., Lewandowsky, S., and Hertwig, R. (2020). Citizens versus the internet: Confronting digital challenges with cognitive tools. *Psychological Science in the Public Interest*, 21(3):103–156.

Lazer, D. M. J., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., Metzger, M. J., Nyhan, B., Pennycook, G., Rothschild, D., Schudson, M., Sloman, S. A., Sunstein, C. R., Thorson, E. A., Watts, D. J., and Zittrain, J. L. (2018). The science of fake news. *Science*, 359(6380):1094–1096.

Lewandowsky, S., Ecker, U. K. H., and Cook, J. (2017). Beyond misinformation: Understanding and coping with the post-truth era. *Journal of Applied Research in Memory and Cognition*, 6(4):353–369.

Matz, S. C., Kosinski, M., Nave, G., and Stillwell, D. J. (2017). Psychological targeting as an effective approach to digital mass persuasion. *Proceedings of the National Academy of Sciences*, 114(48):12714–12719.

Pennycook, G. and Rand, D. G. (2019). Lazy, not biased: Susceptibility to partisan fake news is better explained by lack of reasoning than by motivated reasoning. *Cognition*, 188:39–50.

Pennycook, G. and Rand, D. G. (2021). The psychology of fake news. *Trends in Cognitive Sciences*, 25(5):388–402.

Roozenbeek, J., van der Linden, S., Goldberg, B., Rathje, S., and Lewandowsky, S. (2022). Psychological inoculation improves resilience against misinformation on social media. *Science Advances*, 8(34):eabo6254.

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2):1–36.

Shrout, P. E. and Fleiss, J. L. (1979). Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, 86(2):420–428.